strategy-with-position-management

June 20, 2024

1 Trading Strategy with Position Management on MSFT using Pandas_TA

[1]:
```
import pandas as pd
import numpy as np
import yfinance as yf
import pandas_ta
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

[2]:
```
data = yf.download('MSFT', period='5y')
data.head()
```

```
[*********************100%%**********************]  1 of 1 completed
```

[2]:
```
                  Open        High         Low       Close   Adj Close  \
Date
2019-06-20  137.449997  137.660004  135.720001  136.949997  130.844208
2019-06-21  136.580002  137.729996  136.460007  136.970001  130.863312
2019-06-24  137.000000  138.399994  137.000000  137.779999  131.637222
2019-06-25  137.250000  137.589996  132.729996  133.429993  127.481125
2019-06-26  134.350006  135.740005  133.600006  133.929993  127.958862

              Volume
Date
2019-06-20  33042600
2019-06-21  36727900
2019-06-24  20628800
2019-06-25  33327400
2019-06-26  23657700
```

[3]:
```
data['PVI'] = pandas_ta.pvi(close=data['Adj Close'],
                            volume = data['Volume'],
                            length=20)
```

[4]:
```
data['RSI'] = pandas_ta.rsi(close=data['Adj Close'], length=20)
data['RSI_lag'] = data['RSI'].shift()
data= data.dropna()
data.head()
```

[4]:
```
                  Open        High         Low       Close   Adj Close  \
Date
2019-07-22  137.410004  139.190002  137.330002  138.429993  132.258194
2019-07-23  139.759995  139.990005  138.029999  139.289993  133.079880
2019-07-24  138.899994  140.740005  138.850006  140.720001  134.446121
2019-07-25  140.429993  140.610001  139.320007  140.190002  133.939728
2019-07-26  140.369995  141.679993  140.300003  141.339996  135.038452

              Volume          PVI        RSI    RSI_lag
Date
2019-07-22  25074900   999.759032  57.448799  48.881431
2019-07-23  18034600   999.759032  60.739873  57.448799
2019-07-24  20738300  1005.222583  65.420866  60.739873
2019-07-25  18356900  1005.222583  62.512877  65.420866
2019-07-26  19037600  1010.582249  65.967838  62.512877
```

[5]:
```
data['Signal_1'] = data['PVI'].diff().apply(lambda x:1 if x>1 else(-1 if x<1␣
 ↪else np.nan))
data['Signal_2'] = data.apply(lambda x: 1 if (x['RSI_lag']<50) & (x['RSI']>50)
                              else (-1 if (x['RSI_lag']>50) & (x['RSI']<50) else␣
 ↪np.nan), axis=1)
```

[6]:
```
data.head(1)
```

[6]:
```
                  Open        High         Low       Close   Adj Close  \
Date
2019-07-22  137.410004  139.190002  137.330002  138.429993  132.258194

              Volume         PVI        RSI    RSI_lag  Signal_1  Signal_2
Date
2019-07-22  25074900  999.759032  57.448799  48.881431       NaN       1.0
```

[7]:
```
data['Position'] = data.apply(lambda x:'Buy' if (x['Signal_1'] +␣
 ↪x['Signal_2']==2)
                              else 'Sell' if (x['Signal_1'] + x['Signal_2']==-2)
                              else np.nan, axis=1)
```

[8]:
```
data2 = data[data['Position'].notna()]
```

[9]:
```
data2.head(4)
```

[9]:
```
                 Open        High         Low       Close   Adj Close  \
Date
2019-07-31  140.330002  140.490005  135.080002  136.270004  130.194534
2019-08-02  138.089996  138.320007  135.259995  136.899994  130.796432
2019-08-12  137.070007  137.860001  135.240005  135.789993  129.735947
2019-08-13  136.050003  138.800003  135.000000  138.600006  132.420654

              Volume          PVI        RSI    RSI_lag  Signal_1  Signal_2  \
Date
2019-07-31  38598800  1013.797201  44.939990  60.714103      -1.0      -1.0
2019-08-02  30791600  1014.233698  47.376321  50.838616      -1.0      -1.0
2019-08-12  20484300  1008.901186  46.910538  50.817196      -1.0      -1.0
2019-08-13  25154600  1010.010046  52.532349  46.910538       1.0       1.0

           Position
Date
2019-07-31     Sell
2019-08-02     Sell
2019-08-12     Sell
2019-08-13      Buy
```

[10]:
```python
def strategy_implementation(data, investment):

    equity = investment
    pos = 0
    buy=[]
    sell = []

    for i in range(len(data['Position'])):
        if data['Position'][i] == 'Sell':
            sell.append(1)
            pos -= 1
            equity += data['Adj Close'][i]
            print(f"SELL at price {data['Adj Close'][i]:.2f}. Equity: {equity:.
 ↪2f}.")

        elif data['Position'][i] == 'Buy':
            buy.append(1)
            pos += 1
            equity -= data['Adj Close'][i]
            print(f"BUY at price {data['Adj Close'][i]:.2f}. Equity: {equity:.
 ↪2f}.")

    #Make sure no positions held at the end of the period
    if pos>1:
        stocks_sold = pos
        equity += stocks_sold * data['Adj Close'][i]
```

```
        print(f"SELL at price {data['Adj Close'][i]:.2f}. Equity: {equity:.2f}.␣
↪Stocks sold to close existing position {stocks_sold}.")
        sell.append(1)

    elif pos<1:
        stocks_bought = abs(pos)
        equity -= stocks_bought * data['Adj Close'][i]
        print(f"BUY at price {data['Adj Close'][i]:.2f}. Stocks bought to close␣
↪existing position {stocks_bought}.")
        buy.append(1)

    print('')
    print(f"Number of Trades {len(buy)+len(sell)}.")
    print(f"Number of Buy trades {len(buy)}.")
    print(f"Number of Sell trades {len(sell)}.")
```

[11]:
```
strategy_implementation(data,0)
```

```
SELL at price 130.19. Equity: 130.19.
SELL at price 130.80. Equity: 260.99.
SELL at price 129.74. Equity: 390.73.
BUY at price 132.42. Equity: 258.31.
SELL at price 128.43. Equity: 386.74.
SELL at price 127.87. Equity: 514.61.
SELL at price 130.41. Equity: 645.01.
SELL at price 130.45. Equity: 775.46.
BUY at price 131.83. Equity: 643.63.
SELL at price 130.69. Equity: 774.32.
SELL at price 131.39. Equity: 905.71.
SELL at price 131.44. Equity: 1037.16.
SELL at price 131.72. Equity: 1168.88.
SELL at price 130.72. Equity: 1299.60.
SELL at price 158.67. Equity: 1458.27.
SELL at price 160.37. Equity: 1618.63.
SELL at price 146.71. Equity: 1765.34.
BUY at price 159.40. Equity: 1605.94.
SELL at price 196.70. Equity: 1802.65.
SELL at price 196.48. Equity: 1999.13.
SELL at price 200.94. Equity: 2200.07.
SELL at price 199.90. Equity: 2399.97.
SELL at price 203.67. Equity: 2603.65.
SELL at price 196.50. Equity: 2800.15.
BUY at price 209.79. Equity: 2590.35.
SELL at price 204.58. Equity: 2794.93.
SELL at price 205.18. Equity: 3000.11.
SELL at price 205.88. Equity: 3205.99.
SELL at price 206.32. Equity: 3412.31.
```

```
SELL at price 208.92. Equity: 3621.23.
BUY at price 218.07. Equity: 3403.16.
SELL at price 223.10. Equity: 3626.26.
SELL at price 221.71. Equity: 3847.97.
SELL at price 224.79. Equity: 4072.76.
SELL at price 225.89. Equity: 4298.65.
BUY at price 229.71. Equity: 4068.94.
SELL at price 241.42. Equity: 4310.36.
SELL at price 240.82. Equity: 4551.18.
SELL at price 238.88. Equity: 4790.06.
SELL at price 241.60. Equity: 5031.65.
SELL at price 287.82. Equity: 5319.47.
SELL at price 316.61. Equity: 5636.09.
SELL at price 318.47. Equity: 5954.55.
SELL at price 322.49. Equity: 6277.05.
BUY at price 295.09. Equity: 5981.96.
SELL at price 265.84. Equity: 6247.79.
BUY at price 263.55. Equity: 5984.24.
SELL at price 252.51. Equity: 6236.75.
SELL at price 249.74. Equity: 6486.49.
SELL at price 256.33. Equity: 6742.82.
BUY at price 264.58. Equity: 6478.24.
SELL at price 264.50. Equity: 6742.74.
SELL at price 228.22. Equity: 6970.96.
BUY at price 239.72. Equity: 6731.23.
SELL at price 233.31. Equity: 6964.54.
SELL at price 247.19. Equity: 7211.73.
SELL at price 246.57. Equity: 7458.30.
SELL at price 273.18. Equity: 7731.47.
BUY at price 292.97. Equity: 7438.51.
SELL at price 328.74. Equity: 7767.25.
SELL at price 325.54. Equity: 8092.80.
SELL at price 327.79. Equity: 8420.59.
SELL at price 413.64. Equity: 8834.23.
SELL at price 409.34. Equity: 9243.57.
BUY at price 424.01. Equity: 8819.56.
BUY at price 444.35. Stocks bought to close existing position 41.

Number of Trades 66.
Number of Buy trades 13.
Number of Sell trades 53.
```

[12]:

```
plt.figure(figsize=(20,15))
plt.style.use("bmh")
plt.plot(data['Adj Close'], label='Closing Price', linestyle="-", linewidth=2,␣
 ↪color='blue')
buy = data2[data2['Position']=='Buy']['Adj Close']
```

```
plt.scatter(buy.index, buy, marker="^", color='g', s=400, label='Buy Signal')
sell = data2[data2['Position']=='Sell']['Adj Close']
plt.scatter(sell.index, sell, marker="v", color='r', s=400, label='Sell Signal')
plt.title("MSFT Stock Price and Signals", fontsize=20)
plt.xlabel("Date", fontsize=16)
plt.ylabel("Closing Price", fontsize=16)
plt.legend(loc='upper left', markerscale=0.5,fontsize=15)
plt.show()
```